MA416

Lecture 14 – October 21, 2025

Introduction to Non-Parametric Testing
& Bootstrap Sampling

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Some Tests We Have Explored…

Some of the "parametric" tests we have learned are…

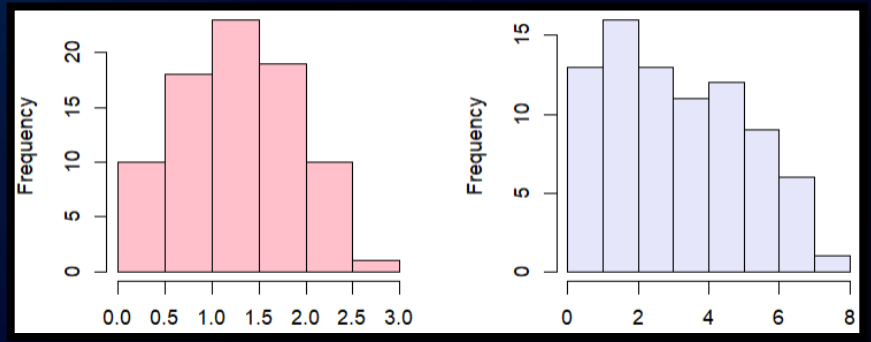
- One-Sample T-test for Means
  - Requires normality (or large sample size for CLT)
- Two-Sample T-test for Mean Differences
  - Requires 2 x normality (or 2 x large sample sizes for 2 x CLT)
- ANOVA
  - Requires g x Normality
- Bartlett
  - Requires g x Normality

Or from previous courses…

- Tests on correlation or least-squares coefficients
  - Requires bivariate normality

```
mydata = read.csv('CompanyPayroll_9.25.csv')
X = mydata$Prior_Experience
Y = mydata$Efficiency_Factor_Obs

par(mfrow=c(1,2))
hist(X, col='pink')
hist(Y, col='lavender')
```




```
source('mynormtest.R') # Nemo built
Nx = mynormtest(X)
Ny = mynormtest(Y)
c(Nx$KS) # returns 0.0581
c(Ny$KS) # returns 0.0889
```

Tests that require the population to follow a specific structure (for example a distribution and/or specific range of parameters) are called "parametric tests".

Bootstrap Sampling

Definition: "Bootstrapping" is a statistical method that estimates the accuracy or variability of a sample statistic by repeatedly resampling with replacement (of the entire sample size) from the original data and recalculating the statistic each time.

Bootstrapping is:

- A resampling technique – it draws many new samples from the observed data.
- A non-parametric method – it doesn't assume the population follows any specific distribution.

Bootstrapping is not:

- A parametric method – it doesn't rely on formulas tied to any distribution.
- A way to create new information – it only reuses what's already in your data; it can't fix bias or bad sampling.

Word of Caution: Bootstrapping works best when your original sample is large and representative.

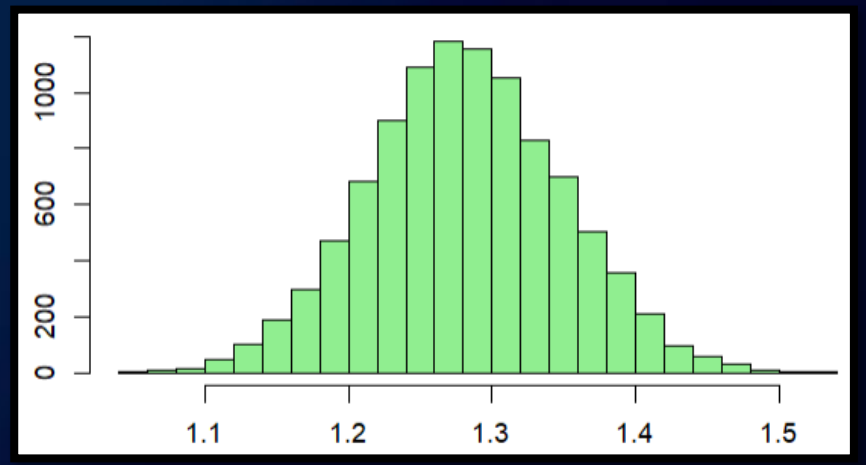
Bootstrap Sampling

Example 1: One-Sample Bootstrap Confidence Intervals

Suppose you want to construct a confidence interval for the sample mean prior experience for a population.

```
set.seed(123)
nboot = 10000
mymean = c()
for(k in 1:nboot){
  sboot = sample(X, length(X), replace=T)
  mymean = c(mymean, mean(sboot))
}

hist(mymean, breaks=20, col='lightgreen')
```




```
alpha = 0.05
cimean = quantile(mymean, c(alpha/2, 1-alpha/2))
cimean
```

```
> cimean
      2.5%      97.5%
1.150364198 1.415435185
```

Bootstrap Sampling

Example 2: Two-Sample Bootstrap Confidence Intervals

Suppose you want to construct a confidence interval for the correlation coefficient between prior experience and efficiency.

```
n = length(X) # or Y

set.seed(123)
nboot = 10000
mycor = c()

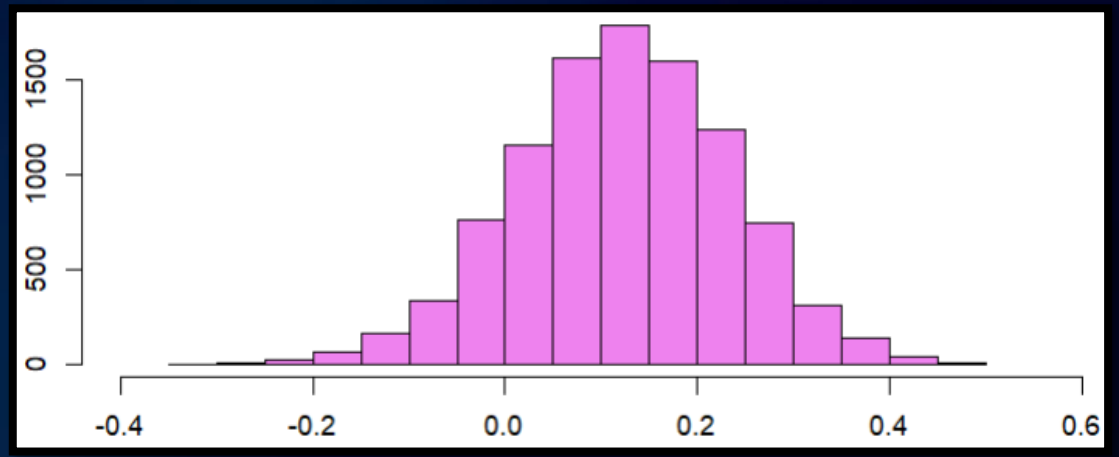
myindex = seq(1, n, 1)

for(k in 1:nboot){
  sboot = sample(myindex, n, replace=T)
  Xboot = X[sboot]
  Yboot = Y[sboot]

  mycor = c(mycor, cor(Xboot, Yboot))

}

hist(mycor, breaks=20, col='violet',
     xlim=c(-0.4, 0.6))
```




```
alpha = 0.03
cirxy = quantile(mycor, c(alpha/2, 1-alpha/2))
cirxy
```

```
> cirxy
      1.5%      98.5%
-0.1291093065  0.3564765238
```

Bootstrap Sampling

Example 3: Two-Sample Bootstrap Confidence Intervals

Suppose you want to construct a confidence interval both the slope and intercept for a least-squares regression line with prior experience as your predictor variable and efficiency as your response variable.

```
set.seed(123)
nboot = 10000
mybhat1 = c()
mybhat0 = c()

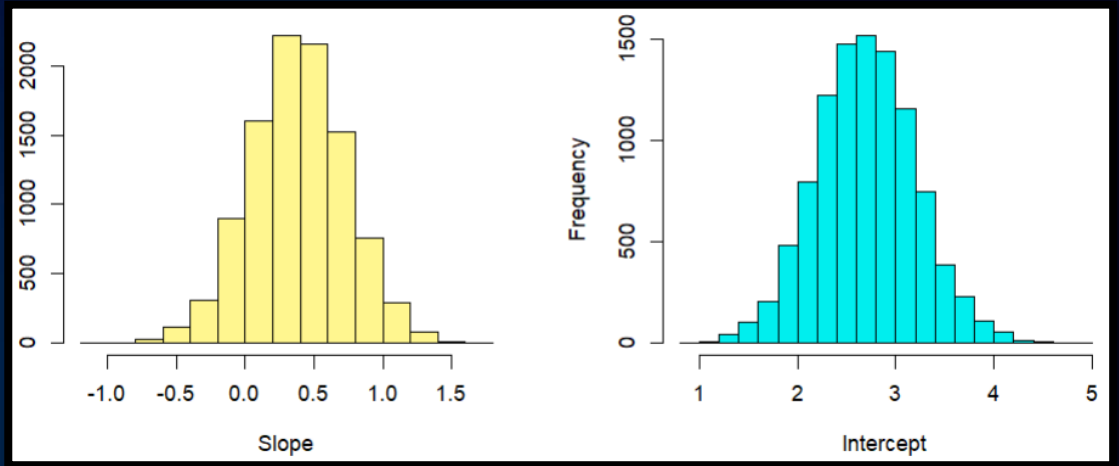
myindex = seq(1, n, 1)

for(k in 1:nboot){
  sboot = sample(myindex, n, replace=T)
  Xboot = X[sboot]
  Yboot = Y[sboot]

  bhat1 = cor(Xboot, Yboot)*sd(Yboot)/sd(Xboot)
  bhat0 = mean(Yboot) - bhat1*mean(Xboot)

  mybhat1 = c(mybhat1, bhat1)
  mybhat0 = c(mybhat0, bhat0)
}

par(mfrow=c(1,2))
hist(mybhat1, breaks=20, col='khaki1', xlab='Slope')
hist(mybhat0, breaks=20, col='cyan2', xlab='Intercept')
```




```
alpha = 9.45/100
cibhat1 = quantile(mybhat1, c(alpha/2, 1-alpha/2))
cibhat0 = quantile(mybhat0, c(alpha/2, 1-alpha/2))
```

```
> rbind(cibhat1, cibhat0)
                  4.725%      95.275%
cibhat1 -0.1956808499 0.9660017347
cibhat0  1.8533651558 3.5601159509
```

Non-Parametric Testing

Some things that can be said about non-parametric testing…

Good Things (for non-parametric tests)

- Few (if any) assumptions
- Robust to outliers
- Can be used on small sample sizes

Bad Things (for non-parametric tests)

- Less statistical power
- Limited information
- Fewer model options

Bad Things (bootstrap sampling)

- It can be computationally expensive
- It may give false confidence with small sample sizes
- It makes outliers appear more common than they are
- It can amplify bias in bad samples
- It assumes your sample reflects the population